METRIC

A-A-1818A
24 May 2005
SUPERSEDING
A-A-1818
24 July 1981

COMMERCIAL ITEM DESCRIPTION

SODIUM METASILICATE, TECHNICAL

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) for sodium metasilicate covers material suitable for various washing, cleaning, and scouring processes, with or without soap as conditions demand, and where a moderately strong alkaline material is desired.

2. CLASSIFICATION. The sodium metasilicate shall be classified by type. The selected type of the product to be supplied shall be one of the following types as specified in the acquisition order (see 7.3(b)):

Type: I - Pentahydrate ($Na_2SiO_3 \cdot 5H_2O$)
II - Anhydrous (Na_2SiO_3)

3. SALIENT CHARACTERISTICS. The sodium metasilicate shall be a uniform product. The product shall conform to the following chemical and physical characteristics:

3.1 Characteristics. The sodium metasilicate, as received, shall conform to the requirements as to chemical composition and particle size prescribed in table I.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 6810

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TABLE I. Chemical and physical requirements.

	Type I		Type II	
	Min	Max	Min	Max
Sodium metasilicate:				
Calculated as $Na_2SiO_3 \cdot 5H_2O$, % by wt.	98.0	--	--	--
Calculated as Na_2SiO_3, % by wt.	--	--	91.5	--
Total alkalinity, as Na_2O, % by wt.	28.5	30.0	50.0	52.0
Total silica as SiO_2, % by wt.	27.5	29.0	45.0	--
Matter insoluble in water, % by wt.	--	0.2	--	0.3
Carbonates as CO_2, % by wt.	--	1.2	--	2.0
Particle size:				
Material passing a 149-micron (No. 100) sieve, % by wt., maximum	--	10.0	--	3.0
Material retained on a 2.00 mm (No. 10) sieve, % by wt., maximum	--	10.0	--	10.0

3.2 Materials. The material shall be sampled and analyzed in accordance with ASTM D 501, "Standard Test Methods of Sampling and Chemical Analysis of Alkaline Detergents".

3.3 Particle size. The particle size shall be sampled and analyzed in accordance with ASTM D 502, "Standard Test Methods of Particle Size of Soaps and Other Detergents".

3.4 Net weight. The sodium metasilicate shall be purchase by net weight but in the event of weight shortage it shall be recognized that sodium metasilicate is subject to a possible gain or loss of weight due to atmospheric or storage condition, or both. Therefore, the calculation of the net weight shall be based upon a product containing 98 percent of sodium metasilicate calculated as $Na_2SiO_3 \cdot 5H_2O$ for Type I or 91.5 percent of sodium metasilicate calculated as Na_2SiO_3 for Type II. The net weight of the material to be paid shall be calculated as follows:

For Type I $W = (R \times T)/98$
For Type II $W = (R \times T')/91.5$

Where:
W = Net weight of the material to be paid for,
R = Net weight of the material as received,
T = total sodium metasilicate calculated as $Na_2SiO_3 \cdot 5H_2O$, and
T' = total sodium metasilicate calculated as Na_2SiO_3

In the event that the calculation results in a figure greater than 100 percent, the seller waives the right to invoice the purchaser for any amount in excess of 100 percent.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

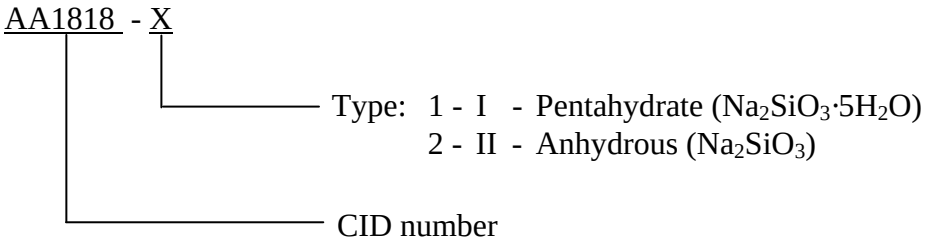
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(c)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



Example of reference part number: AA1818 - 1 indicates Type I sodium metasilicate

7.2 Sources of documents.

7.2.1 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.2.2 ASTM standards. Copies of ASTM standards may be obtained from the ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

7.3 Ordering data. The acquisition order should specify the following information:

a. CID document number, revision, and CID PIN.
b. Type (see 2).
c. Preservation, packing, and marking (see 6.1).

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7.4 National stock numbers (NSNs). The following is a list of the NSNs assigned that correspond to this CID. This list may not be indicative of all possible NSNs associated with this document.

NSN	Type
6810-00-240-2121	I
6810-00-664-7062	II

7.5 Subject term (key word) listing.

Anhydrous
Pentahydrate
Alkaline detergents

MILITARY INTERESTS:

Custodians:
Army - MR
Air Force - 68
NIMA - MP

Review Activities:
Army - EA
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS3

(Project 6810-2005-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at http://assist.daps.dla.mil/.

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